Git & Github

Install git.

- Installed Git on macOS using xcode-select --install and verified the installation with git --version.

1.
```
sharafkamaal@192 devops-techiehorizon % git --version
git version 2.50.1 (Apple Git-155)
sharafkamaal@192 devops-techiehorizon % xcode-select --install
xcode-select: note: Command line tools are already installed. Use "Software Update" in System Settings or the softwareupdate command
line interface to install updates
sharafkamaal@192 devops-techiehorizon %
```

2. Create a repo in github with README.md and .ignore file.

- **Steps to Create a GitHub Repository with README.md and .gitignore**
- Log in to GitHub.

- Click the + icon in the top-right corner and select New repository.

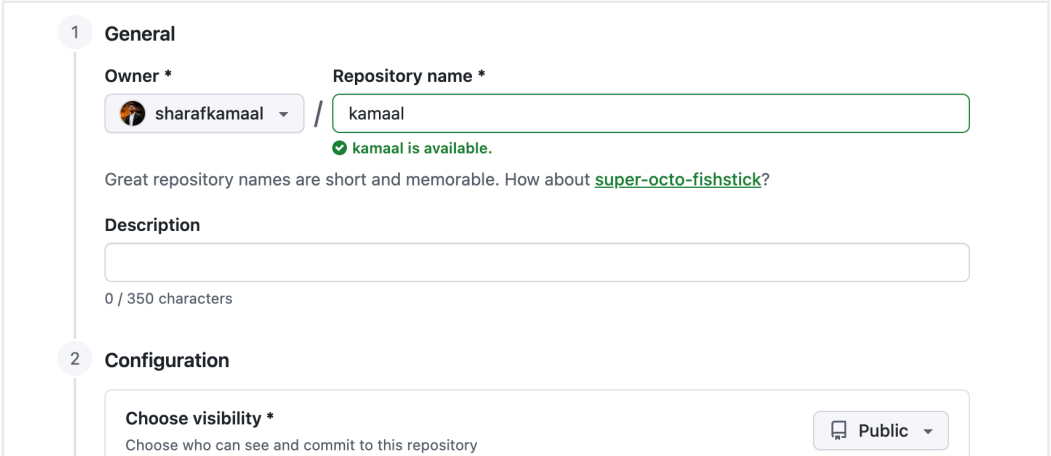
- Enter the repository name.

- Select Public or Private.

- Check Add a README file.

- Under Add .gitignore, select the required template (e.g., Node).

- Click Create repository.



3.Clone the created repo to local.

1. Copy the repository URL from GitHub by clicking Code → HTTPS → Copy URL.
2. Open the Terminal on your MacBook.
3. Navigate to the folder where you want to store the repository.
4. Run the command:

```
git clone https://github.com/USERNAME/REPOSITORY.git
```

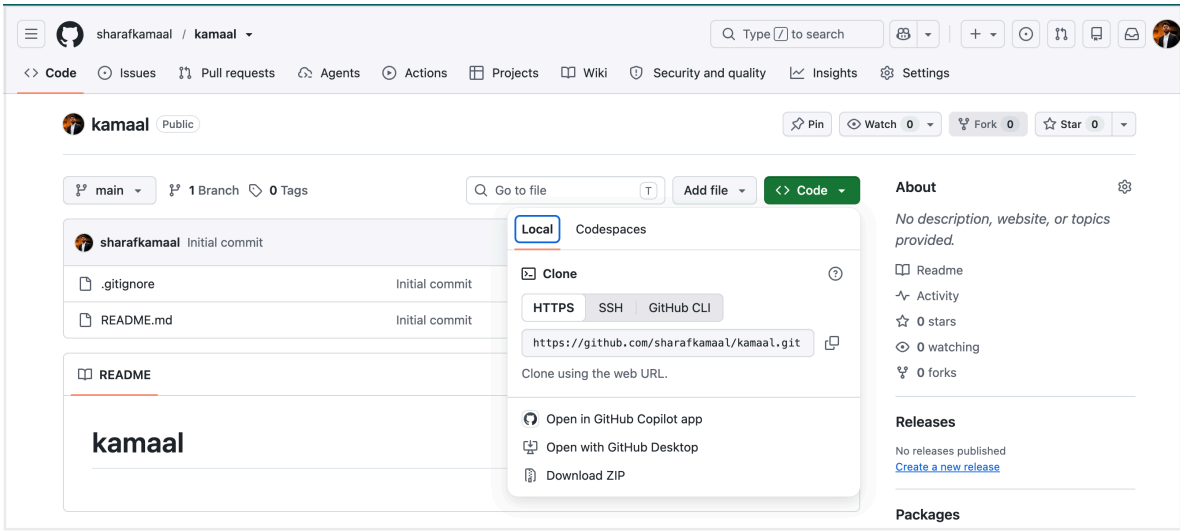
5. Navigate into the cloned repository:

```
cd REPOSITORY
```

6. Verify the files:

```
ls -la
```

You should see README.md, .gitignore, and the .git directory.

1. 


```
sharafkamaal@192 devops-techiehorizon % git clone https://github.com/sharafkamaal/kamaal123.git
Cloning into 'kamaal123'...

remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
sharafkamaal@192 devops-techiehorizon %
sharafkamaal@192 devops-techiehorizon %
```

4.Create two files in local repo.

1. Navigate to your local repository using cd REPOSITORY.
2. Create two files using touch file1.txt file2.txt.
3. Verify the files using ls -l.

4.

```
sharafkamaal@192 devops-techiehorizon % mkdir kamaal123
mkdir: kamaal123: File exists
sharafkamaal@192 devops-techiehorizon % cd kamaal123
sharafkamaal@192 kamaal123 % touch file1.txt file2.txt
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       README.md
sharafkamaal@192 kamaal123 %
```

5. Commit two files and push to central Repository.

```
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git init
Reinitialized existing Git repository in /Users/sharafkamaal/devops-techiehorizon/kamaal123/.git/
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   file1.txt
        new file:   file2.txt
```

6.

```
sharafkamaal@192 kamaal123 % git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 318 bytes | 318.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sharafkamaal/kamaal123.git
   2c23cac..d716abc  main -> main
branch 'main' set up to track 'origin/main'.
sharafkamaal@192 kamaal123 %
```

6.Create a branch in local and create a sample file and push to central.

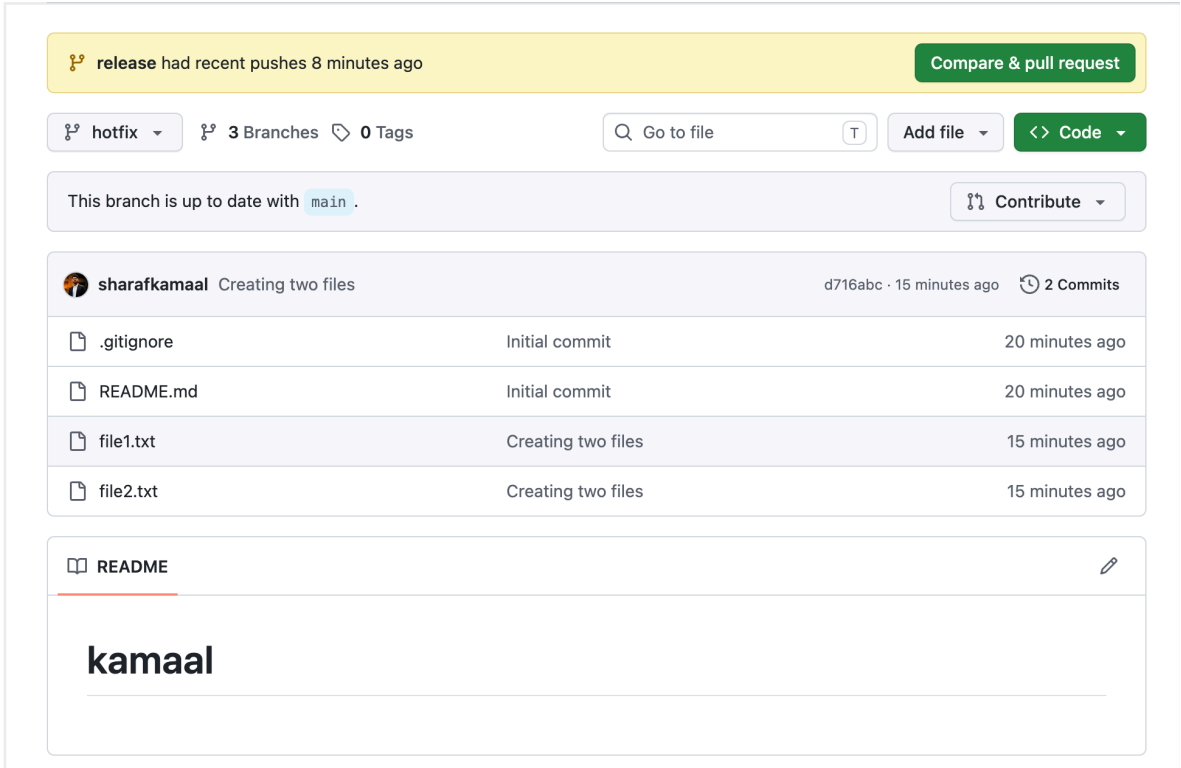
```
branch 'main' set up to track 'origin/main'.
sharafkamaal@192 kamaal123 % git branch --list
* main
sharafkamaal@192 kamaal123 % git branch release
sharafkamaal@192 kamaal123 % git branch --list
* main
  release
sharafkamaal@192 kamaal123 % git checkout release
Switched to branch 'release'
sharafkamaal@192 kamaal123 % git branch --list
  main
* release
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       README.md
sharafkamaal@192 kamaal123 % touch sample.txt
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       README.md       sample.txt
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git commit -m "created a release branch and sample file"
[release 5d3579a] created a release branch and sample file
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 sample.txt
sharafkamaal@192 kamaal123 % git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
sharafkamaal@192 kamaal123 %
```

1.

```
sharafkamaal@192 kamaal123 % git push -u origin release
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 245 bytes | 245.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'release' on GitHub by visiting:
remote:      https://github.com/sharafkamaal/kamaal123/pull/new/release
remote:
To https://github.com/sharafkamaal/kamaal123.git
 * [new branch]      release -> release
branch 'release' set up to track 'origin/release'.
sharafkamaal@192 kamaal123 %
```

7.Create a branch in github and clone that to local.
Open your repository on GitHub.
Click the branch dropdown (usually showing main).
Enter a new branch name, such as dev, and click Create branch: dev.
Copy the repository URL by clicking Code → HTTPS → Copy URL.
Open Terminal and navigate to the folder where you want to clone the repository.
Clone the new branch using git clone -b dev https://github.com/USERNAME/REPOSITORY.git.
Navigate into the cloned repository using cd REPOSITORY.
Verify the current branch using git branch.

1.

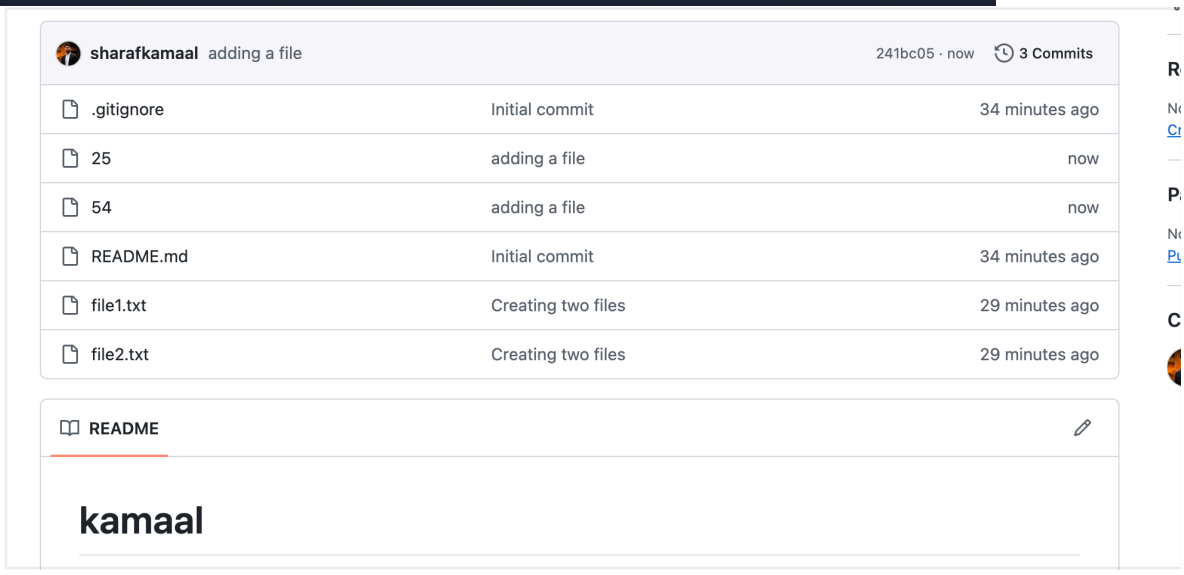



```
sharafkamaal@192 kamaal123 % git clone -b hotfix https://github.com/sharafkamaal/kamaal123.git
Cloning into 'kamaal123'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 2), reused 5 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (2/2), done.
sharafkamaal@192 kamaal123 % cd hotfix
cd: no such file or directory: hotfix
```

3. Merge the created branch with master in git local.
```
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       kamaal123       README.md       sample.txt
sharafkamaal@192 kamaal123 % touch  1 2 3 4
```
4.

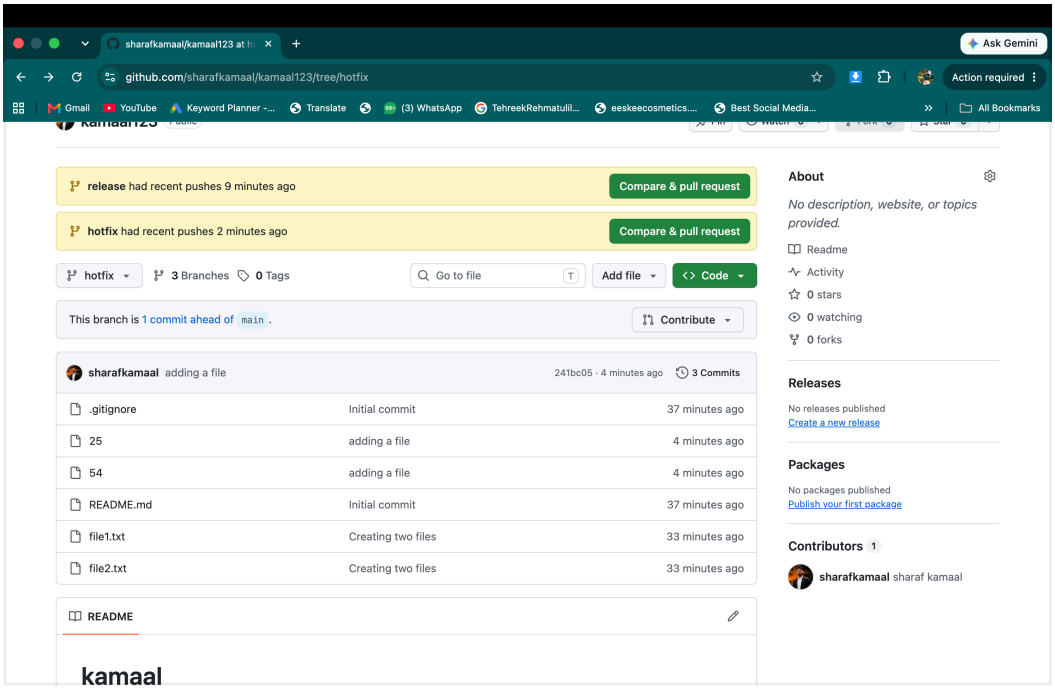
1.
```
sharafkamaal@192 kamaal123 % cd kamaal123
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       README.md
sharafkamaal@192 kamaal123 % touch  54 25
sharafkamaal@192 kamaal123 % ls
25              54              file1.txt       file2.txt       README.md
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git commit -m "adding a file"
[hotfix 241bc05] adding a file
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 25
 create mode 100644 54
sharafkamaal@192 kamaal123 % git push -u origin hotfix
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 235 bytes | 235.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sharafkamaal/kamaal123.git
   d716abc..241bc05  hotfix -> hotfix
branch 'hotfix' set up to track 'origin/hotfix'.
sharafkamaal@192 kamaal123 % git checkout main
branch 'main' set up to track 'origin/main'.
Switched to a new branch 'main'
sharafkamaal@192 kamaal123 % git switch main
Already on 'main'
Your branch is up to date with 'origin/main'.
sharafkamaal@192 kamaal123 % git merge hotfix
Updating d716abc..241bc05
Fast-forward
 25 | 0
 54 | 0
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 25
 create mode 100644 54
sharafkamaal@192 kamaal123 %
```

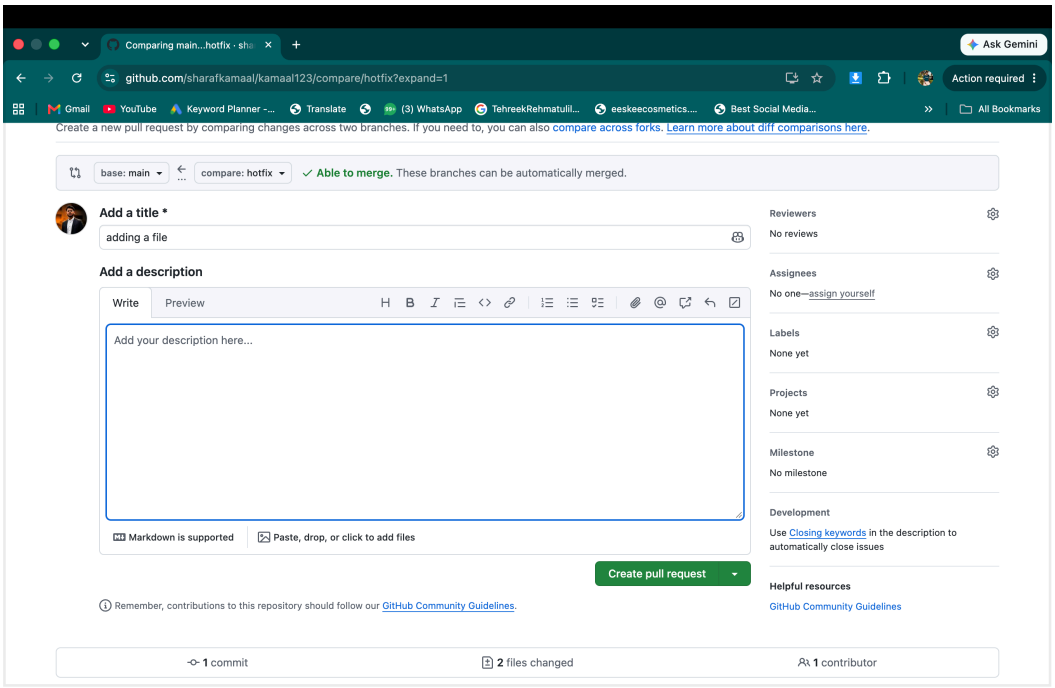
2.

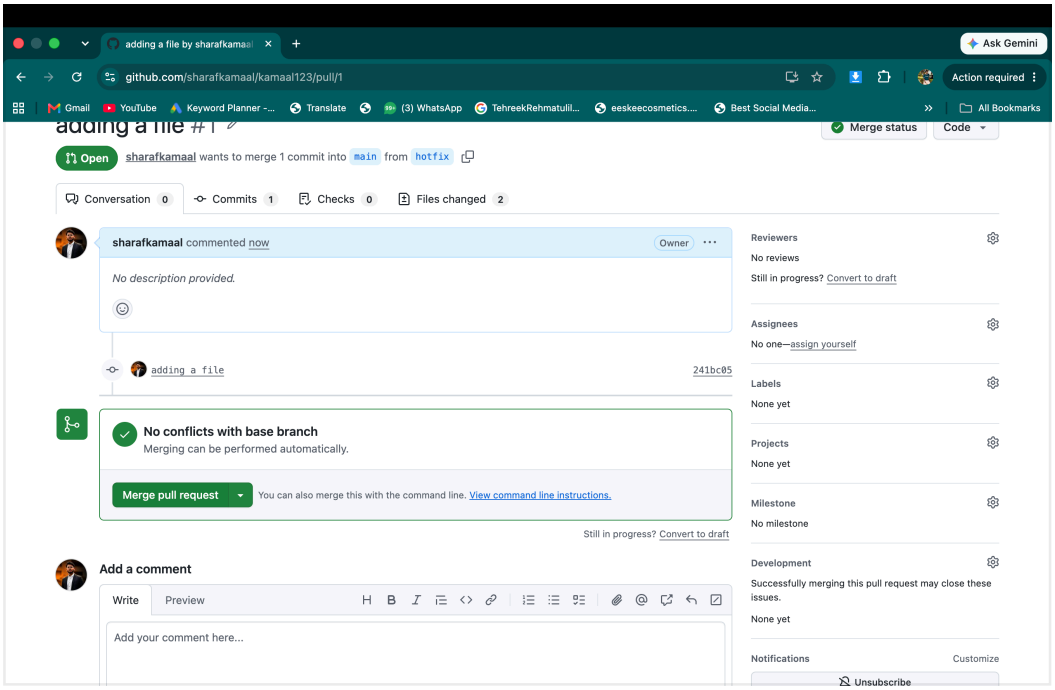


8.Merge the created branch with master in github by sending a pull request.
Open your repository on GitHub.
Click Pull requests → New pull request.
Select main (or master) as the base branch.

Select your created branch, such as dev, as the compare branch.
Click Create pull request.
Enter a title and description, then click Create pull request.
Review the changes and click Merge pull request.
Click Confirm merge.
Verify that the changes from dev are now available in main (or master).

1. 





9.Create a file in local and send that to branch in github.
Open Terminal and navigate to your local repository using cd REPOSITORY.
Switch to the branch you created using git switch dev.
Create a new file using touch file3.txt.
Add some content to the file using echo "Hello GitHub" > file3.txt.
Check the changes using git status.
Stage the file using git add file3.txt.
Commit the file using git commit -m "Add file3".
Push the changes to GitHub using git push origin dev.
Open your GitHub repository and select the dev branch to verify that file3.txt is available.

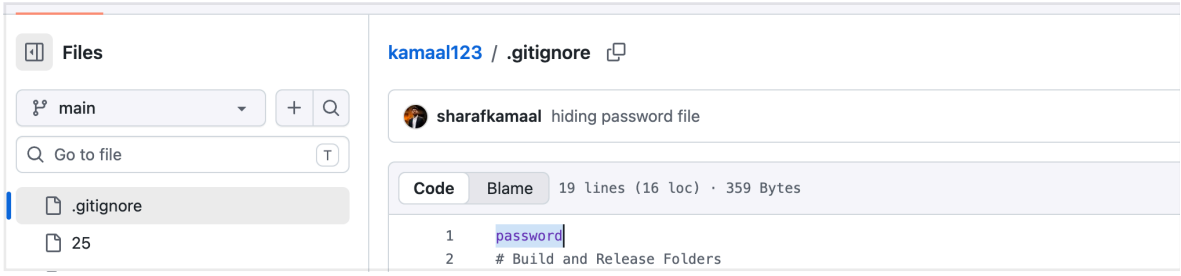
1.
```
[sharafkamaal@192 kamaal123 % ls
 file1.txt       file2.txt       README.md
[sharafkamaal@192 kamaal123 % touch  54 25
[sharafkamaal@192 kamaal123 % ls
 25              54              file1.txt       file2.txt       README.md
[sharafkamaal@192 kamaal123 % git add .
[sharafkamaal@192 kamaal123 % git commit -m "adding a file"
[hotfix 241bc05] adding a file
 2 files changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 25
 create mode 100644 54
[sharafkamaal@192 kamaal123 % git push -u origin hotfix
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 10 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 235 bytes | 235.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sharafkamaal/kamaal123.git
   d716abc..241bc05  hotfix -> hotfix
branch 'hotfix' set up to track 'origin/hotfix'.
```

10.Clone only a branch from github to local.

1.
```
sharafkamaal@192 kamaal123 % git clone -b hotfix https://github.com/sharafkamaal/kamaal123.git
Cloning into 'kamaal123'...

remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 11 (delta 3), reused 6 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), done.
Resolving deltas: 100% (3/3), done.
sharafkamaal@192 kamaal123 %
sharafkamaal@192 kamaal123 % cd kamaal123
sharafkamaal@192 kamaal123 % ls
file1.txt       file2.txt       README.md
```

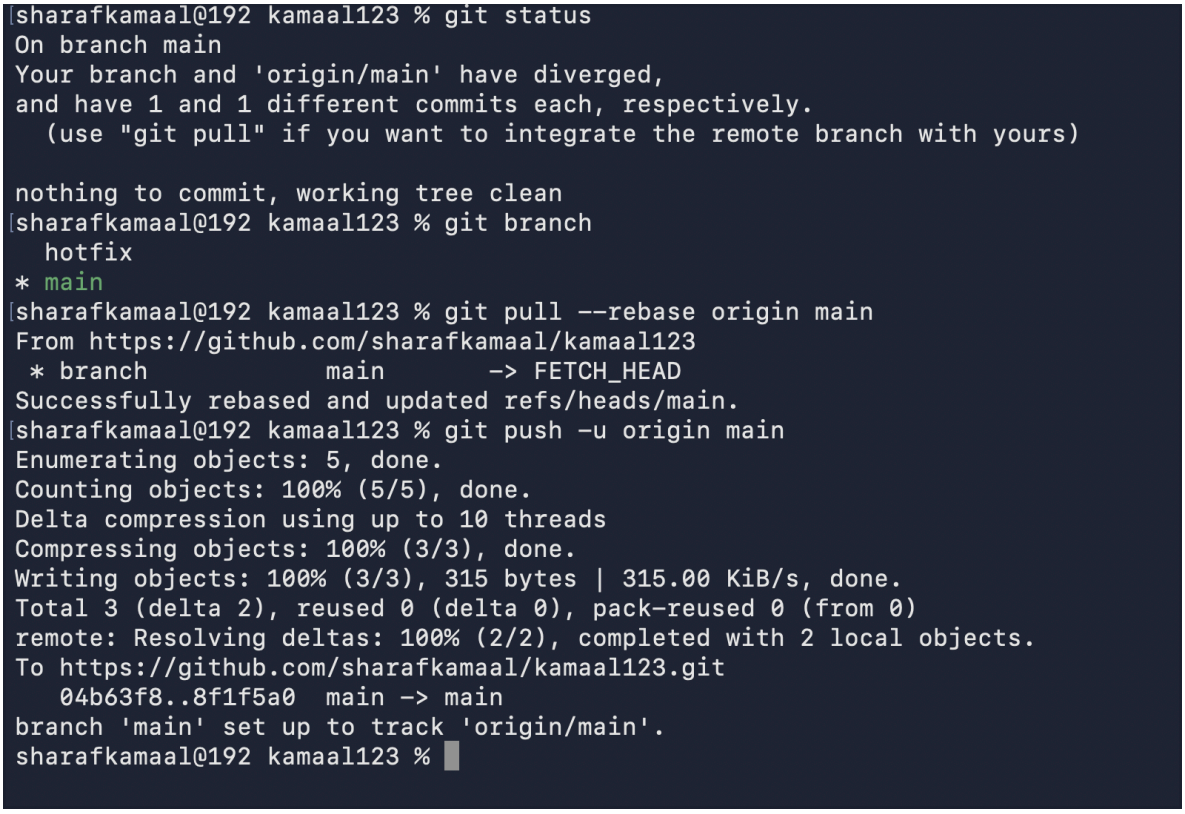
11.Create a file with all passwords and make that untrackable with git.
Open Terminal and navigate to your local repository using cd REPOSITORY.
Switch to the branch you created using git switch dev.
Create a new file using touch file3.txt.
Add some content to the file using echo "Hello GitHub" > file3.txt.
Check the changes using git status.
Stage the file using git add file3.txt.
Commit the file using git commit -m "Add file3".
Push the changes to GitHub using git push origin dev.
Open your GitHub repository and select the dev branch to verify that file3.txt is available.

1.



```
sharafkamaal@192 kamaal123 % vi password
sharafkamaal@192 kamaal123 % ls
25              54              file1.txt       file2.txt       password        README.md
sharafkamaal@192 kamaal123 % vi .gitignore
sharafkamaal@192 kamaal123 % ls
25              54              file1.txt       file2.txt       password        README.md
sharafkamaal@192 kamaal123 % vi . gitignore
2 files to edit
sharafkamaal@192 kamaal123 % ls
25              54              file1.txt       file2.txt       password        README.md
sharafkamaal@192 kamaal123 % ls -a
.               .git            25              file1.txt       password
..              .gitignore      54              file2.txt       README.md
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   .gitignore

no changes added to commit (use "git add" and/or "git commit -a")
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git commit -m "hiding password file"
[main c302fda] hiding password file
 1 file changed, 1 insertion(+)
```



12.Make a commit and make that commit reset without savings changes.

1. Open Terminal and navigate to your local repository using cd REPOSITORY.

2. Create a file using echo "Test content" > reset.txt.

3. Stage the file using git add reset.txt.

4. Commit the changes using git commit -m "Add reset file".

5. Reset the latest commit and discard all its changes using git reset --hard HEAD~1.

6. Verify the result using git status.

7. Confirm that reset.txt is deleted and the latest commit has been removed from your local branch.

Note: git reset --hard HEAD~1 permanently discards the latest commit and its

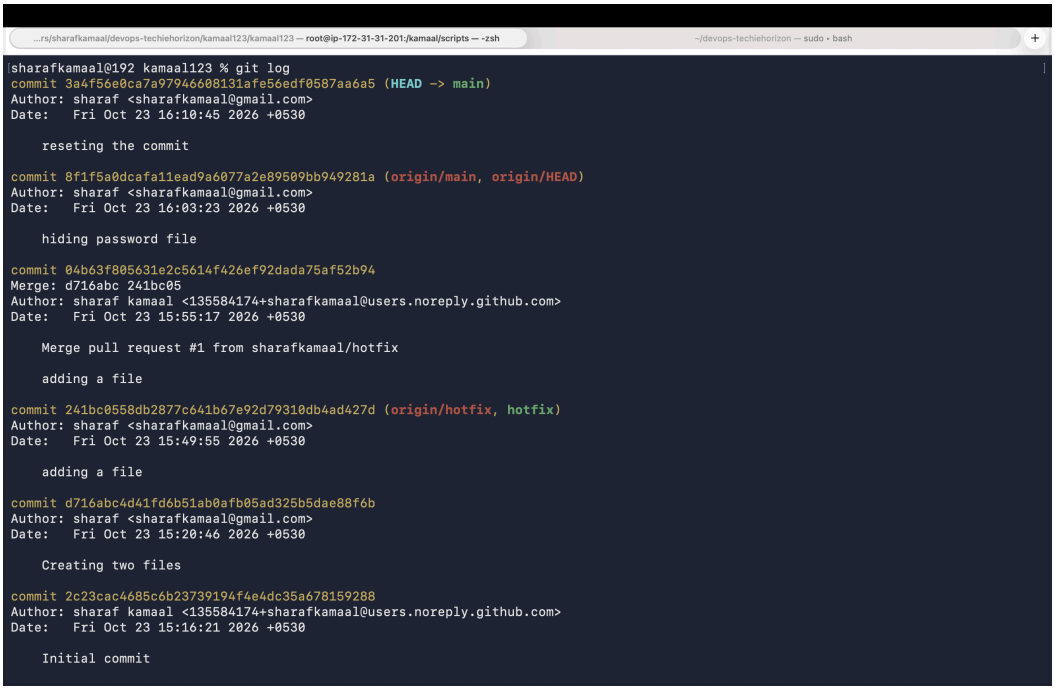
tracked changes from your current branch.

1.

```
sharafkamaal@192 kamaal123 % touch commit
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        commit

nothing added to commit but untracked files present (use "git add" to track)
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git commit -m "reseting the commit"
[main 3a4f56e] reseting the commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 commit
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)
```



```
sharafkamaal@192 kamaal123 % git log
commit 3a4f56e0ca7a97946608131afe56edf0587aa6a5 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:10:45 2026 +0530

    reseting the commit

commit 8f1f5a0dcafa11ead9a6077a2e89509bb949281a (origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:03:23 2026 +0530

    hiding password file

commit 04b63f805631e2c5614f426ef92dada75af52b94
Merge: d716abc 241bc05
Author: sharaf kamaal <135584174+sharafkamaal@users.noreply.github.com>
Date:   Fri Oct 23 15:55:17 2026 +0530

    Merge pull request #1 from sharafkamaal/hotfix

    adding a file

commit 241bc0558db2877c641b67e92d79310db4ad427d (origin/hotfix, hotfix)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 15:49:55 2026 +0530

    adding a file

commit d716abc4d41fd6b51ab0afb05ad325b5dae88f6b
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 15:20:46 2026 +0530

    Creating two files

commit 2c23cac4685c6b23739194f4e4dc35a678159288
Author: sharaf kamaal <135584174+sharafkamaal@users.noreply.github.com>
Date:   Fri Oct 23 15:16:21 2026 +0530

    Initial commit
```

```
sharafkamaal@192 kamaal123 % git reset  3a4f56
```

13.Revert a commited commit to the older version.

Note: git revert creates a new commit that undoes the selected commit while preserving the existing commit history.

1.

```
sharafkamaal@192 kamaal123 % git revert 241bc0
[main 5f0e43d] Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
 2 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 25
 delete mode 100644 54
sharafkamaal@192 kamaal123 %
```

```
Revert "adding a file"
"doing some changes"
This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
```

14.Push a file to stash without savings the changes and work on another file.

1. Open Terminal and navigate to your local repository using cd REPOSITORY.

2. Create two files using touch file1.txt file2.txt.

3. Add content to the first file using echo "First file changes" > file1.txt.

4. Stash the changes without committing them using git stash push -m "Save file1 changes".

5. Verify that the changes are stashed using git stash list.

6. Add content to the second file using echo "Second file changes" > file2.txt.

7. Stage and commit the second file using git add file2.txt and git commit -m "Add file2 changes".

8. Verify the commit using git log --oneline.

9. Restore the stashed changes when ready using git stash pop.

Note: git stash temporarily saves your uncommitted changes so you can work on another task without committing them.

```
sharafkamaal@192 kamaal123 % vi alpha
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        alpha

nothing added to commit but untracked files present (use "git add" to track)
sharafkamaal@192 kamaal123 % git add .
sharafkamaal@192 kamaal123 % git stash
Saved working directory and index state WIP on main: 5f0e43d Revert "adding a file" "doing some changes" This reverts commit 241bc055
8db2877c641b67e92d79310db4ad427d.
sharafkamaal@192 kamaal123 %
sharafkamaal@192 kamaal123 % git stash list
stash@{0}: WIP on main: 5f0e43d Revert "adding a file" "doing some changes" This reverts commit 241bc0558db2877c641b67e92d79310db4ad4
27d.
sharafkamaal@192 kamaal123 % git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
```

15.Undo the stash file and start working on that again.

1. Open Terminal and navigate to your local repository using cd REPOSITORY.

2. Check the stashed changes using git stash list.

3. Restore the latest stashed changes and remove them from the stash list using git stash pop.

4. Check the restored changes using git status.

5. Open the restored file and continue editing it.

6. Stage the changes using git add file1.txt.

7. Commit the changes using git commit -m "Continue work on file1".

8. Verify the commit using git log --oneline.

Note: git stash pop restores your stashed changes so you can continue working on them.

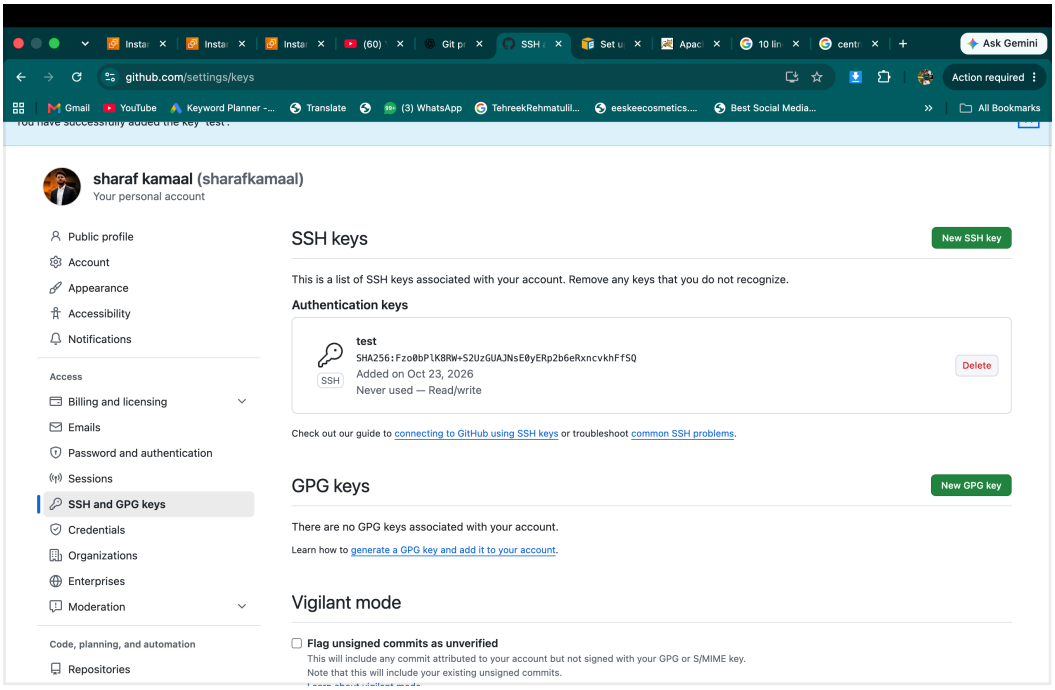
1.
```
sharafkamaal@192 kamaal123 % git stash pop
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   alpha

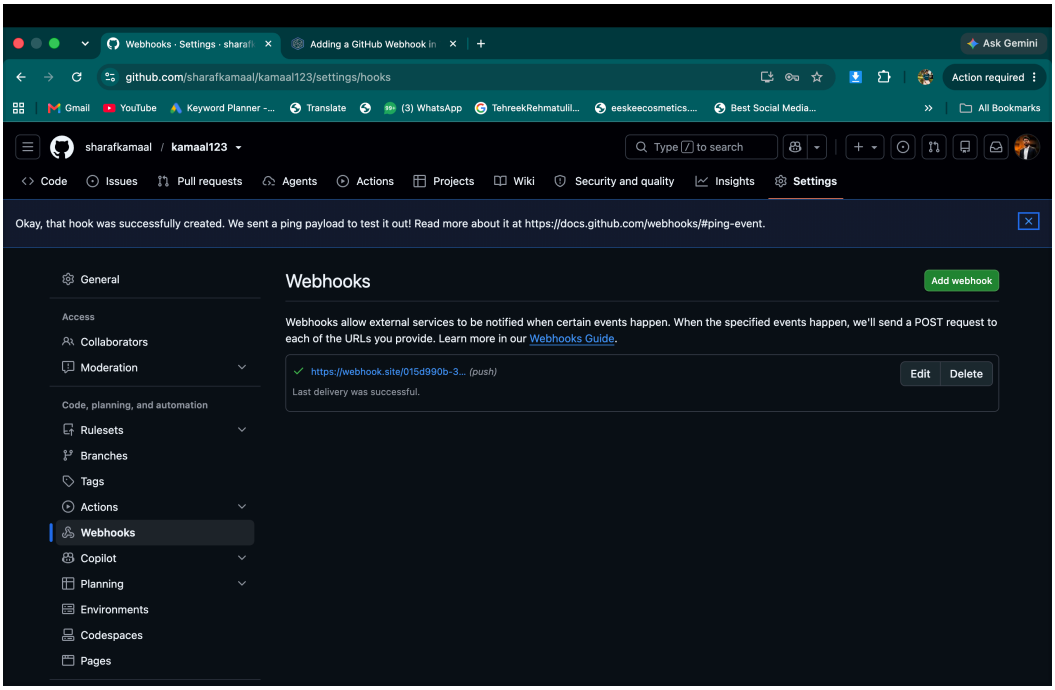
Dropped refs/stash@{0} (01482d49dadb88365711c6a815b55d44ebbf6ca8)
sharafkamaal@192 kamaal123 %
```

16.Generate a ssh-keygen and configure into github.

```
sharafkamaal@192 kamaal123 % ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/Users/sharafkamaal/.ssh/id_ed25519):
/Users/sharafkamaal/.ssh/id_ed25519 already exists.
Overwrite (y/n)? y
Enter passphrase for "/Users/sharafkamaal/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /Users/sharafkamaal/.ssh/id_ed25519
Your public key has been saved in /Users/sharafkamaal/.ssh/id_ed25519.pub
```



17.Configure webhooks to github.
Open your repository on GitHub.
Click Settings → Webhooks.
Click Add webhook.
Enter your webhook endpoint URL in the Payload URL field (e.g., http://YOUR_SERVER_IP:8080/github-webhook/ for Jenkins).
Select application/json as the Content type.
Add a secret if your webhook receiver is configured to validate it.
Under Which events would you like to trigger this webhook?, select Just the push event.
Ensure Active is checked.
Click Add webhook.
Open the webhook details and check Recent Deliveries to verify whether GitHub successfully delivered the request.



18.Basic understanding of .git file.

1 Navigate to your local Git repository using cd REPOSITORY.

1. Display hidden files using ls -la.
2. Locate the .git directory, which stores Git repository metadata and version-control information.
3. View the current branch reference using cat .git/HEAD.
4. Check repository configuration using cat .git/config.
5. List stored Git objects using ls .git/objects.
6. View local branch references using ls .git/refs/heads.
7. Check the staging index using git status.

1.

```
sharafkamaal@192 kamaal123 % git init
Reinitialized existing Git repository in /Users/sharafkamaal/devops-techiehorizon/kamaal123/kamaal123/.git/
sharafkamaal@192 kamaal123 % ls
alpha           file1.txt       file2.txt       password        README.md
sharafkamaal@192 kamaal123 % ls -la
total 32
drwxr-xr-x   9 sharafkamaal  staff  288 23 Oct 16:50 .
drwxr-xr-x  14 sharafkamaal  staff  448 23 Oct 17:47 ..
drwxr-xr-x  16 sharafkamaal  staff  512 23 Oct 18:36 .git
-rw-r--r--   1 sharafkamaal  staff  359 23 Oct 16:06 .gitignore
-rw-r--r--   1 sharafkamaal  staff   51 23 Oct 16:50 alpha
-rw-r--r--   1 sharafkamaal  staff    0 23 Oct 15:49 file1.txt
-rw-r--r--   1 sharafkamaal  staff    0 23 Oct 15:49 file2.txt
-rw-r--r--   1 sharafkamaal  staff   33 23 Oct 15:59 password
-rw-r--r--   1 sharafkamaal  staff    8 23 Oct 15:49 README.md
sharafkamaal@192 kamaal123 %
```

19.Check all the logs of git.

1. Open Terminal and navigate to your local repository using cd REPOSITORY.
2. View the complete Git commit history using git log.
3. View a compact list of commits using git log --oneline.
4. View the latest 5 commits using git log -5 --oneline.
5. View commits with author names and dates using git log --pretty=format:"%h - %an - %ad - %s" --date=short.
6. View the changes introduced by each commit using git log -p.
7. View the history of a specific file using git log -- file1.txt.
8. View the current repository status using git status.

```
        log
sharafkamaal@192 kamaal123 % git log
commit 5f0e43dfdcc49e53b71d229a5b7a41b0df946fe3 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:31:00 2026 +0530

    Revert "adding a file"
    "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

commit 8f1f5a0dcafa11ead9a6077a2e89509bb949281a (origin/main, origin/HEAD)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:03:23 2026 +0530

    hiding password file

commit 04b63f805631e2c5614f426ef92dada75af52b94
Merge: d716abc 241bc05
Author: sharaf kamaal <135584174+sharafkamaal@users.noreply.github.com>
Date:   Fri Oct 23 15:55:17 2026 +0530

    Merge pull request #1 from sharafkamaal/hotfix

    adding a file

commit 241bc0558db2877c641b67e92d79310db4ad427d (origin/hotfix, hotfix)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 15:49:55 2026 +0530

    adding a file

commit d716abc4d41fd6b51ab0afb05ad325b5dae88f6b
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 15:20:46 2026 +0530

    Creating two files

commit 2c23cac4685c6b23739194f4e4dc35a678159288
```

20. Rename the commit message.

```
...vops-techiehorizon/kamaal123/kamaal123 — root@ip-172-31-31-201:/kamaal/scripts — vi • git commit --amend    ~/devops-techiehorizon — sudo • bash

Revert "adding a file"
"doing some changes"
This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

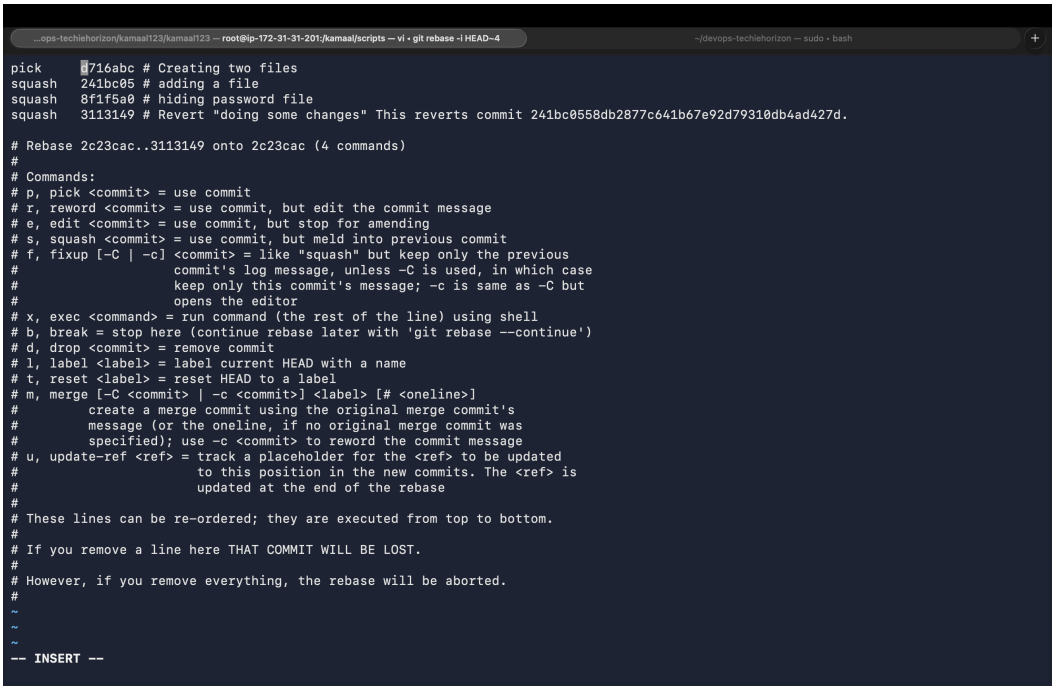
# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# Date:      Fri Oct 23 16:31:00 2026 +0530
#
# On branch main
# Your branch is ahead of 'origin/main' by 1 commit.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#       deleted:    25
#       deleted:    54
#       new file:   alpha
#
~
~
~
~
~
~
```

```
commit 3113149b198922dacaef08704578b33128d0245a (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 16:31:00 2026 +0530

    Revert "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.
```

22.Merge multiple commits into single commit.

1.




```
[sharafkamaal@192 kamaal123 % git rebase -i HEAD~4
[detached HEAD 9d940b4] Creating two files
 Date: Fri Oct 23 15:20:46 2026 +0530
 4 files changed, 20 insertions(+)
 create mode 100644 alpha
 create mode 100644 file1.txt
 create mode 100644 file2.txt
Successfully rebased and updated refs/heads/main.
```

```
sharafkamaal@192 kamaal123 % git log
commit 9d940b49d1441e8ca1acec92fd1bfacd2d3b9292 (HEAD -> main)
Author: sharaf <sharafkamaal@gmail.com>
Date:   Fri Oct 23 15:20:46 2026 +0530

    Creating two files

    adding a file

    hiding password file

    Revert "doing some changes"
    This reverts commit 241bc0558db2877c641b67e92d79310db4ad427d.

commit 2c23cac4685c6b23739194f4e4dc35a678159288
Author: sharaf kamaal <135584174+sharafkamaal@users.noreply.github.com>
Date:   Fri Oct 23 15:16:21 2026 +0530

    Initial commit
sharafkamaal@192 kamaal123 %
```